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Assignment 1: screenshot

Task	Model	Classification (Success/Failure)	Observation (What actually happened?)	Why did this happen? (Architectural Reason)
Generation	BERT	Failure	Generated meaningless / random text and warnings; not coherent sentence generation.	BERT is an encoder only model and is not trained for next-token prediction .
	RoBERTa	Failure	Similar to BERT — cannot generate meaningful sentences.	RoBERTa is also an encoder-only Transformer , trained only with MLM.
	BART	Success	Produced readable and structured generated text.	BART is an encoder-decoder (seq2seq) model trained for text generation.
Fill-Mask	BERT	Success	Correctly predicted masked words such as “create”, “generate”.	Trained using Masked Language Modeling (MLM) .
	RoBERTa	Success	Predicted accurate masked tokens with high confidence.	Improved MLM training with more data and no NSP task.
	BART	Partial Success	Can fill masks but predictions less accurate than BERT/RoBERTa.	BART is optimized for denoising sequence reconstruction , not pure MLM.

Task	Model	Classification (Success/Failure)	Observation (What actually happened?)	Why did this happen? (Architectural Reason)
QA	BERT	Success	Returned an answer like: <i>"as hallucinations, bias, and deepfakes"</i> .	Fine-tuned versions of BERT perform extractive QA effectively.
	RoBERTa	Success	Produced accurate span-based answers from context.	Strong contextual embeddings improve answer span detection.
	BART	Partial Success	Returned short or incomplete answers (low score ≈ 0.016).	BART is mainly designed for generative QA , not extractive QA without fine-tuning.

Assignment 2: Screenshots

```
from transformers import pipeline
```

```
story_generator = pipeline(
    "text-generation",
    model="gpt2"
)
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in your Google Colab and restart you
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
warnings.warn(
config.json: 100% ██████████ 665/665 [00:00<00:00, 10.7KB/s]
model.safetensors: 100% ██████████ 548M/548M [00:08<00:00, 54.4MB/s]
generation_config.json: 100% ██████████ 124/124 [00:00<00:00, 1.85kB/s]
tokenizer_config.json: 100% ██████████ 26.0/26.0 [00:00<00:00, 861B/s]
vocab.json: 100% ██████████ 1.04M/1.04M [00:00<00:00, 8.99MB/s]
merges.txt: 100% ██████████ 456k/456k [00:00<00:00, 9.90MB/s]
tokenizer.json: 100% ██████████ 1.36M/1.36M [00:00<00:00, 15.9MB/s]
Device set to use cpu
```

```
prompt = "The knight entered the dark cave"
```

```
story = story_generator(  
    prompt,  
    max_length=120,  
    do_sample=True,  
    temperature=0.9,  
    top_k=50,  
    top_p=0.95  
)  
  
print(story[0]["generated_text"])
```

... Truncation was not explicitly activated but `max\_length` is provided a specific value, please use `truncation=True` to explicitly truncate examples to max length. Defaulting setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation. Both `max\_new\_tokens` (=256) and `max\_length` (=120) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. ([https://huggingface.co/docs/transformers/main/en/generation_strategies#truncate](#))
The knight entered the dark cave, and a huge flame erupted from the chest. The knight could feel the burning sensation.

When he felt his body burn like fire, his vision disappeared. The light was flickering brightly, and the knight's heart was filled with fear, and he felt the force of his sword. At this moment, a black figure jumped out of the cave and began moving towards the knight. It moved backwards in time, and the knight's sword cut off the red blade.

"I'm in shock, you're-?"

The figure stood up from its hiding place. A small smile appeared on its face.

"I'm not in this cave. I want to return to my village! I want to go on a quest with you. I want to get back to the village.."

"I'm not in this cave. I want to return to my village! I want to go on a quest with you. I want to get back to the village.."

... There was no point in saying this, and he had no chance of telling them anything. The knight thought to himself, "If this guy's still here, then I really shouldn't be so bad. After saying that, the knight's face lit up like fire. At this moment, the knight's eyes were filled with a murderous intent, and his eyes flashed with light. This is the knight

```
def generate_story(starting_sentence):  
    result = story_generator(  
        starting_sentence,  
        max_length=120,  
        do_sample=True,  
        temperature=0.9,  
        top_k=50,  
        top_p=0.95  
    )  
    return result[0]["generated_text"]  
  
user_input = input("Enter a starting sentence: ")  
print("\nGenerated Story:\n")  
print(generate_story(user_input))
```

... Enter a starting sentence: Once upon a time there lived a old woman
Setting `pad\_token\_id` to `eos\_token\_id`:50256 for open-end generation. Both `max\_new\_tokens` (=256) and `max\_length` (=120) seem to have been set. `max\_new\_tokens` will take precedence. Please refer to the documentation for more information. ([https://huggingface.co/docs/transformers/main/en/generation_strategies#truncate](#))

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Generated Story:

Once upon a time there lived a old woman, as a young child of the old, who in the course of their lives had become a great minister for the new religion, a man of wisdom, and this is to say, that before the time of Christ the Old Testament religion was under the dominion of a vast and powerful empire, which had taken the form of an empire that would. The Old Testament religion was founded on the principle that it is possible to do the Lord's work by